

Human Computer Interaction

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“A funny quote by a professor”

- Prof. Dr. Professor Name, YEAR

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Summary of the course materials

Contents

1	Introduction	3
2	User Centered Design	4
2.1	Introduction	4
2.2	Design Process	4

1 Introduction

This summary is likely to end up being quite a bit shorter than my other summaries due to the limited scope of this course, since its primary focus is the project and most of the theory covered is to be applied there, too.

2 User Centered Design

2.1 Introduction

To answer what User-Centred Design (UCD) is, we first need to establish what it *isn't*:

- UCD doesn't mean talking *to* the users and asking what they want (that's pretty much a user study)
- UCD doesn't mean testing the usability of the design in the end
- UCD is not just for user interfaces or software (it can also be for Programming Languages)
- UCD doesn't guarantee a perfect design (duh)

So, what is it then? It is characterized by:

1. **An early focus on users and tasks** (get a psychology degree first)
2. **Empirical measurement** (Come up with some sort of measurement of performance)
3. **Iterative Design** (Identify and fix problems)

That all is very obvious and the hardest part for Computer Scientists of course is the first property. The second property is typically done via user studies, where you keep track of a lot of data on how users interact with your application. Don't shy away from storing way more details than you think you could possibly use, it will make it much easier to evaluate things later on. (Can you tell that I am speaking from experience?)

That brings us to a formal definition from "The Design of Everyday Things" by Donald A. Norman (2013), which also includes famous examples such as the [Norman Door](#).

User-Centred Design

Definition 2.1.1

User-Centered Design (UCD) is an iterative design process in which designers focus on the users and their needs in each phase of the design process. UCD involves researching user behaviors, needs, and experiences and incorporating user feedback through multiple iterations of prototyping and testing to ensure usability, accessibility, and satisfaction. The goal is to develop solutions that align with real user needs rather than assumptions about those needs.

2.2 Design Process